

Test Plan

Document Information

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Module	Login
Document Name	Test Plan
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Prepared By	Anandaroop Maitra
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Revision History

Version	Date	Author	Description
1.0	04 Jul 2026	Anandaroop Maitra	Initial Test Plan

Reference Documents

Version	Document Type	Document Name
1.0	BRD	Business Requirement Document
Current	Application	OrangeHRM Demo
1.0	Test Plan	Login Module Test Plan

Review and Approval

Name	Role	Status
Anandaroop Maitra	QA Engineer	Prepared
Self-Review	Reviewer	Completed

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1. Introduction

This Test Plan defines the overall testing strategy, scope, objectives, resources, schedule, and deliverables for the Login module of the OrangeHRM Demo application. The objective is to verify that the Login functionality meets the business requirements defined in the Business Requirement Document and performs reliably under expected operating conditions.

1.1. Acceptance Criteria

The Login module will be considered ready for release when all of the following conditions are satisfied:

- Users can log in successfully using valid credentials.
- Invalid login attempts display appropriate validation messages.
- Mandatory field validations work as expected.
- The Forgot Password functionality is accessible from the Login page.
- Successful authentication redirects users to the Dashboard.
- Logout terminates the user session successfully.
- Protected pages cannot be accessed after logout.
- No Critical or High severity defects remain open.
- All planned test cases have been executed.
- Stakeholder approval has been obtained for release.

2. Scope

2.1. In Scope

Functional Coverage

The following Login module functionalities are included in the scope of testing:

- Username field validation
- Password field validation
- Login button functionality
- Forgot Password link accessibility
- Successful login with valid credentials
- Unsuccessful login with invalid credentials
- Mandatory field validations
- Error message validation
- Session creation after successful login
- Logout functionality
- Session termination after logout

- Access restriction to protected pages after logout

Form Factor Coverage

Form Factor	Applicable
Desktop	Yes
Mobile Web	No
Android	No
iOS	No

Browser Coverage

Browser	Version
Google Chrome	Latest Stable
Microsoft Edge	Latest Stable
Mozilla Firefox	Latest Stable

Operating System Coverage

Operating System	Version
Windows	Windows 11

Test Types Included

Test Type	Included
Functional Testing	Yes
UI Testing	Yes
Regression Testing	Yes
Validation Testing	Yes
Smoke Testing	Yes
Compatibility Testing	Yes
Negative Testing	Yes

Environment Coverage

Environment	Applicable
Development	No
QA	Yes
UAT	No
Production	No

Exemptions

The following activities are excluded from this test cycle:

- Performance Testing
- Security Testing
- Load Testing
- API Testing
- Database Testing
- Mobile Application Testing
- Accessibility Testing
- Localization Testing

2.2. Out Scope

The following modules are outside the scope of this Test Plan:

- Employee Management
- Recruitment
- Leave Management
- Time Management
- Performance Module
- Directory Module
- Maintenance Module
- Claim Module
- Admin Module
- Dashboard widgets other than post login verification

3. Test Strategy

3.1. Test Type Checklist

Test Type	Applicable	Remarks
Functional Testing	Yes	Core functionality validation
Smoke Testing	Yes	Verify build stability before detailed testing
Regression Testing	Yes	Ensure existing functionality is unaffected
Compatibility Testing	Yes	Validate across supported browsers
UI Testing	Yes	Verify user interface elements
Negative Testing	Yes	Validate invalid input scenarios
Security Testing	No	Outside project scope
Performance Testing	No	Outside project scope
API/Service Testing	No	Login API not in scope
Latency Testing	No	Outside project scope
Memory Leak Testing	No	Not applicable for this project
Installation/Uninstallation Testing	No	Web application
Low Network Testing	No	Outside project scope
Localization Testing	No	Single language testing
Accessibility Testing	No	Outside project scope
Bug Bash	No	Not planned

3.2. Test Strategy Detail

Functional Testing

Functional testing will verify that all Login module features work according to the Business Requirement Document. Positive and negative scenarios will be executed to validate expected behavior.

Smoke Testing

Smoke testing will be performed before detailed testing to verify that the Login page loads successfully, and the application is stable for further testing.

Regression Testing

Regression testing will be performed after defect fixes to ensure that previously working functionality has not been affected.

Compatibility Testing

Testing will be executed on supported desktop browsers:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

UI Testing

UI testing will verify:

- Field alignment
- Labels
- Buttons
- Hyperlinks
- Error messages
- Page layout

Negative Testing

Negative testing will verify the application's behavior when invalid or unexpected input is provided.

Examples:

- Blank Username
- Blank Password
- Invalid Username
- Invalid Password
- Both fields blank

3.3. Test Coverage

Operating System Coverage

Operating System	Version
Windows 11	Latest Stable
Windows 10	Latest Stable

Browser Coverage

Browser	Version
Google Chrome	Latest Stable
Microsoft Edge	Latest Stable
Mozilla Firefox	Latest Stable

Screen Resolution Coverage

Priority	Resolution
P0	1920 × 1080
P1	1366 × 768

3.4. Automation Plan

3.4.1. API Automation

Automation Strategy / Approach

API automation is not planned for this project as the testing scope is limited to UI based functional validation of the Login module.

Call Outs

Not Applicable.

Framework Used

Not Applicable.

Packages Used

Not Applicable.

APIs Planned for Automation

Not Applicable.

Test Cases Eligible for Automation

Not Applicable.

3.4.2. UI Automation

Automation Strategy / Approach

UI automation will be implemented using Selenium WebDriver with Java and TestNG. The automation framework will follow the Page Object Model (POM) design pattern to improve maintainability, reusability, and scalability. Stable and repetitive Login scenarios will be automated to support smoke and regression testing.

Call Outs

- Dynamic waits will be implemented using Explicit Waits.
- Test data will be maintained separately from the test scripts.
- Cross browser execution will be supported through TestNG configuration.
- Reports will be generated using Extent Reports.

Framework Used

Item	Details
Framework Type	Hybrid Framework
Design Pattern	Page Object Model (POM)
Language	Java
Automation Tool	Selenium WebDriver
Test Framework	TestNG
Build Tool	Maven
Reporting	Extent Reports
Version Control	Git & GitHub

Functionalities Planned for Automation

- Valid Login
- Invalid Login
- Blank Username validation

- Blank Password validation
- Invalid Username validation
- Invalid Password validation
- Logout
- Dashboard verification after successful login

Test Cases Eligible for Automation

Test Case ID	Scenario
TC_LOGIN_001	Verify Login using valid credentials
TC_LOGIN_002	Verify Login using invalid username
TC_LOGIN_003	Verify Login using invalid password
TC_LOGIN_004	Verify blank Username validation
TC_LOGIN_005	Verify blank Password validation
TC_LOGIN_006	Verify successful Logout

3.4.3. Latency Automation

Automation Strategy / Approach

Not Applicable.

Performance and latency testing are outside the scope of the current project.

3.5. Test Environment

Test Environment

Item	Details
Application Name	OrangeHRM Demo
Module	Login
Application Type	Web Application
Test Environment	Demo Environment
URL	https://opensource-demo.orangehrmlive.com
Browser	Google Chrome, Microsoft Edge, Mozilla Firefox
Operating System	Windows 11
Internet Connection	Stable Broadband

Hardware Configuration

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB or above
Storage	SSD
Display Resolution	1920 × 1080

Software Requirements

Software	Version
Windows	11
Google Chrome	Latest Stable
Microsoft Edge	Latest Stable
Mozilla Firefox	Latest Stable
Microsoft Office	2021 or Microsoft 365
Git	Latest Stable
Java	JDK 21
Maven	Latest Stable
IntelliJ IDEA	Community Edition

3.6. Test Data

Test Data

Test Data	Purpose
Valid Username	Verify successful login
Valid Password	Verify successful login
Invalid Username	Verify invalid login validation
Invalid Password	Verify invalid login validation
Blank Username	Verify mandatory field validation
Blank Password	Verify mandatory field validation

Invalid Username and Password	Verify authentication failure
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Test Accounts

Account Type	Purpose
Admin User	Functional testing of Login and Logout
Invalid User	Negative testing

3.7. Test Scenarios

UI Test Scenarios

Scenario ID	Test Scenario
TS001	Verify successful login using valid credentials
TS002	Verify login with invalid username
TS003	Verify login with invalid password
TS004	Verify login with both invalid username and password
TS005	Verify mandatory field validation
TS006	Verify Forgot Password functionality
TS007	Verify successful logout
TS008	Verify access restriction after logout

3.8. Test Cases

Detailed test cases are maintained in a separate Test Case document.

Reference:

docs/04_Test_Cases/Test_Cases_OrangeHRM_Login.xlsx

3.9. Defects

All defects identified during testing will be logged with the following information:

- Defect ID
- Summary
- Description

- Severity
- Priority
- Environment
- Steps to Reproduce
- Expected Result
- Actual Result
- Status
- Screenshots (if applicable)

Detailed defect reports are maintained separately in the Bug Reports document.

4. Assumptions and Dependencies

Assumptions

- The OrangeHRM Demo application remains available throughout the testing cycle.
- Test user credentials are available.
- Internet connectivity remains stable.
- Supported browsers are installed on the test machine.
- The application build is stable before testing begins.

Dependencies

- Approved Business Requirement Document.
- Approved Test Plan.
- Stable test environment.
- Availability of test credentials.
- Availability of supported browsers.

5. Risks and Contingencies

Risks and Mitigation Plan

Risk	Impact	Mitigation Plan
Test environment unavailable	High	Coordinate with the application owner and resume testing once available.
Test credentials become invalid	High	Maintain backup test accounts and request updated credentials promptly.

Requirement changes during testing	Medium	Review the impact, update test cases, and perform regression testing if required.
Browser compatibility issues	Medium	Execute tests on all supported browsers and report inconsistencies.
Internet connectivity issues	Low	Ensure a stable network connection before execution and rerun affected test cases if necessary.
Defect fixes delayed	Medium	Prioritize critical defects and adjust the test schedule if required.

6. Test Deliverables

Test Deliverables

Deliverable	Description
Business Requirement Document (BRD)	Defines business requirements for the Login module
Test Plan	Defines testing strategy and scope
Test Scenarios	High level testing scenarios
Test Cases	Detailed test cases for execution
Requirement Traceability Matrix (RTM)	Maps requirements to test cases
Test Execution Report	Execution status of all test cases
Bug Report	Details of identified defects
Test Summary Report	Final QA report and testing outcome

7. Dev Build Plan

Dev Build Plan

Build Version	ETA	Notes
Demo Application	Continuous	Public demo environment used for testing

8. QA Entry Criteria

Testing will begin only when all the following conditions are met:

- Business Requirement Document is reviewed and approved.
- Test Plan is reviewed and approved.
- Test cases are prepared and reviewed.
- Test environment is available.
- Test user credentials are available.
- Supported browsers are installed.
- Application is stable for testing.

9. QA Exit Criteria

Testing will be considered complete when all the following conditions are satisfied:

- All planned test cases have been executed.
- All Critical and High severity defects are closed.
- Regression testing has been completed.
- Test execution results have been documented.
- Test Summary Report has been prepared.
- Stakeholder approval has been obtained.

10. Open Questions

No open questions at the time of Test Plan preparation.

11. Effort Estimation

Task	Owner	Estimated Effort
Requirement Analysis	Anandaroop Maitra	2 Hours
Test Planning	Anandaroop Maitra	3 Hours
Test Scenario Design	Anandaroop Maitra	2 Hours
Test Case Design	Anandaroop Maitra	6 Hours
Test Data Preparation	Anandaroop Maitra	1 Hour
Test Execution	Anandaroop Maitra	5 Hours
Defect Reporting	Anandaroop Maitra	2 Hours
Regression Testing	Anandaroop Maitra	2 Hours
Test Summary Report	Anandaroop Maitra	1 Hour